STATS403 Final Exam

Spring'25 Session 4

NAME: ____

NetID: ____

Time: Due by noon, March 7, 2023

Total points: 200pt

NOTE: Please submit a scanned copy of your handwritten exam on canvas.

Problem 1. (30pt)

TRUE or FALSE. For each statement, determine whether it is True or False. In case it is False, please also state your reason.

1. Let A be a fully-connected network (FCN). The numbers of neurons in each layer are (a, b, d), respectively. Suppose we insert a hidden layer in A to get another FCN called B whose numbers of neurons are (a, b, c, d), respectively. Suppose there is no constraint on the parameters. Then B is more prone to overfitting.

2. The neural tangent kernel we discussed in class considers the case where the width of a neural network approaches infinity, but does not explain why we need to make the architecture "deep" in practice.

3. When training neural networks, Adam is more accurate than SGD.

Problem 2. (30 points)

You are designing a Large Language Model (LLM) based on the Transformer architecture. Answer the following questions:

1. Given a token embedding matrix $\mathbf{X} \in \mathbb{R}^{4\times 8}$ (4 tokens, each of dimension 8), describe how to employ a self-attention process to produce *Query*, *Key*, and *Value* matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V}$.

2. Explain why *masked attention* is used during LLM training for next-token prediction. Under the same setting of Part 1, write out the mask matrix.

3. What are potential benefits of using encoder-only and decoder-only LLMs, respectively? You only need to state one benefit for each model.

Problem 3. (25pt)

Suppose you are training a binary classifier. You have a data matrix $\mathbf{X} \in \mathbb{R}^{n \times m}$ with the label vector $\boldsymbol{y}_0$ and implement a neural network with only one hidden layer as follows:

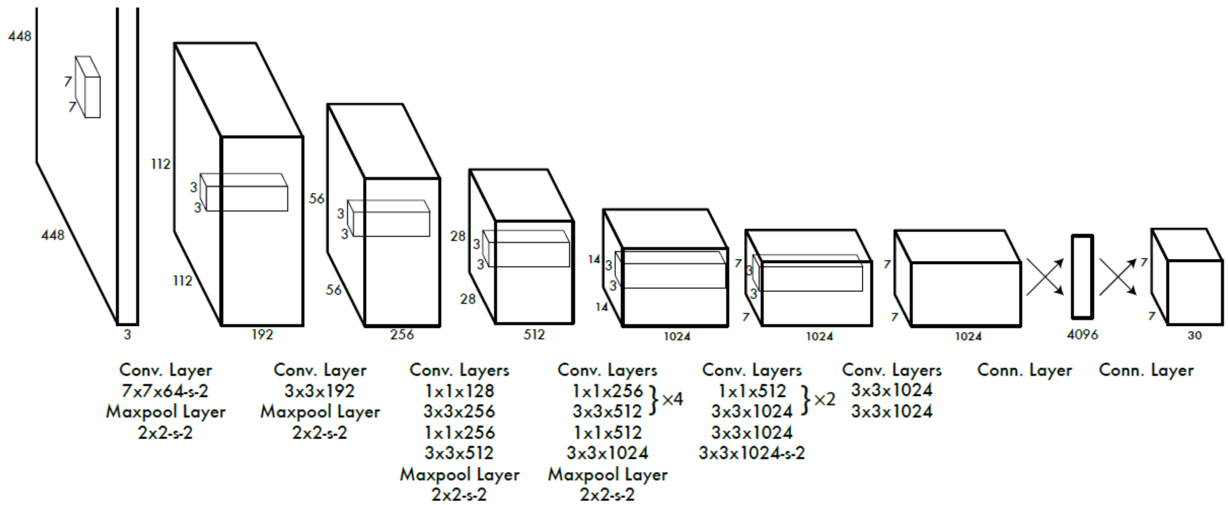
$$\begin{aligned}
\mathbf{Z} &= \sigma(\mathbf{W}_1 \mathbf{X} + \boldsymbol{b}_1) \\
\boldsymbol{y} &= \sigma(\mathbf{W}_2 \mathbf{Z} + \boldsymbol{b}_2) \\
\mathcal{L} &= \mathrm{CrossEntropy}(\boldsymbol{y}, \boldsymbol{y}_0)
\end{aligned}$$

where σ is the nonlinear activation function and $\mathcal{L}$ is the loss function. Here, "+" needs to be understood naturally as bias addition applied to all data points. Suppose the hidden layer has k neurons.

1. What is the shape of $\boldsymbol{y}$?
2. What is the number of parameters in the neural network?
3. What is a possible choice of σ?
4. Explicitly write "CrossEntropy" as a function (using mathematical operations).
5. Derive the gradient of $\mathcal{L}$ with respect to the parameters.

Problem 4. (25pt)

The following figure shows the convolutional neural network (CNN) structure used in the original YOLO (You Only Look Once) model. In this figure, "Conv. Layer" denotes a convolutional layer, while "Conn. Layer" denotes a fully-connected layer.



1. How many convolutional layers (excluding maxpool layers and fully-connected layers) does this CNN have?
2. Explain the meaning of "2×2 - s - 2" in the maxpool layers.
3. For each image, in the first layer of the CNN, what is the shape of the tensor after applying the convolutional layer but before applying the maxpool layer?
4. How many parameters are there in the fully-connected layers?
5. Explain why there are 30 channels in the final output tensor.

Problem 5. (30pt)

The Maximum Mean Discrepancy (MMD) between two domain samples $\mathbf{X}_s, \mathbf{X}_t \subset \mathbb{R}^D$ is defined to be

$$\mathrm{MMD}(\mathbf{X}_s, \mathbf{X}_t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(\boldsymbol{x}_i^s) - \frac{1}{n_t} \sum_{i=1}^{n_t} \phi(\boldsymbol{x}_i^t) \right\|_{\mathcal{H}} ,$$

where $\mathcal{H}$ is a real-valued reproducing kernel Hilbert space (RKHS) with a kernel k such that $k(\boldsymbol{x}, \boldsymbol{x}') = \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle_{\mathcal{H}}$. Define the $(n_s + n_t)$-by-$(n_s + n_t)$ matrix $\mathbf{K}$ to be

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{s,s} & \mathbf{K}_{s,t} \\ \mathbf{K}_{s,t}^\top & \mathbf{K}_{t,t} \end{bmatrix}$$

in which the (i, j)-th entry of $\mathbf{K}_{u,v}$ is given by $k(\boldsymbol{x}_i^u, \boldsymbol{x}_j^v)$ (where $u, v \in \{s, t\}$). Also define a matrix $\mathbf{L}$ with the same shape as $\mathbf{K}$ to be

$$\mathbf{L} = \begin{bmatrix} \mathbf{L}_{s,s} & \mathbf{L}_{s,t} \\ \mathbf{L}_{s,t}^\top & \mathbf{L}_{t,t} \end{bmatrix}$$

where $\mathbf{L}_{s,s}$ is a block matrix (with the same shape as $\mathbf{K}_{s,s}$) whose entries are all $\frac{1}{n_s^2}$, $\mathbf{L}_{t,t}$ is a block matrix whose entries are all $\frac{1}{n_t^2}$ and $\mathbf{L}_{s,t}$ is a block matrix whose entries are all $-\frac{1}{n_s n_t}$.

1. Prove: $\mathrm{MMD}^2(\mathbf{X}_s, \mathbf{X}_t) = \mathrm{tr}(\mathbf{KL})$.
2. In general, for two distributions P and Q we define $\mathrm{MMD}(P, Q) = \|\mathbb{E}_{\mathbf{x}\sim P}(\phi(\mathbf{x})) - \mathbb{E}_{\mathbf{x}\sim Q}(\phi(\mathbf{x}))\|_{\mathcal{H}}$.
   (a) Prove: For any $f \in \mathcal{H}$,
   $$\|f\|_{\mathcal{H}} = \sup_{\|g\|_{\mathcal{H}} \leq 1} \langle g, f \rangle_{\mathcal{H}} .$$
   (b) Prove:
   $$\mathrm{MMD}(P, Q) = \sup_{\|f\|_{\mathcal{H}} \leq 1} \mathbb{E}_{\mathbf{x}\sim P}(f(\mathbf{x})) - \mathbb{E}_{\mathbf{x}\sim Q}(f(\mathbf{x})) .$$

Problem 6. (30pt)

The total variation (TV) distance between two probability distributions $\mathcal{D}$, $\mathcal{D}'$ is defined by

$$\mathrm{TV}(\mathcal{D}, \mathcal{D}') = 2 \sup_{E \in \mathcal{B}} |\mathbb{P}_{\mathcal{D}}(E) - \mathbb{P}_{\mathcal{D}'}(E)| \ ,$$

where $\mathcal{B}$ is a set of events measurable by $\mathcal{D}$ and $\mathcal{D}'$, $\mathbb{P}_{\mathcal{D}}(E)$ is the probability of the event E under the distribution $\mathcal{D}$ and similarly is $\mathbb{P}_{\mathcal{D}'}(E)$ defined. (Remark: in literature you might find the TV distance defined without the factor "2".)

1. Prove: if $\mathcal{B}$ is the collection of all subsets of a finite set Ω, then

$$\mathrm{TV}(\mathcal{D}, \mathcal{D}') = \sum_{x \in \Omega} |\mathbb{P}_{\mathcal{D}}(x) - \mathbb{P}_{\mathcal{D}'}(x)| \ .$$

2. Recall that in the theory of transfer learning, the $\mathcal{H}$-divergence is defined by

$$d_{\mathcal{H}}(\mathcal{D}, \mathcal{D}') = 2 \sup_{h \in \mathcal{H}} |\mathbb{P}_{\mathcal{D}}(I(h)) - \mathbb{P}_{\mathcal{D}'}(I(h))| \ ,$$

where $x \in I(h) \Leftrightarrow h(x) = 1$. Can $d_{\mathcal{H}}$ be written in the form of a TV distance? If yes, specify $\mathcal{B}$.

3. Assume $\mathbb{P}_{\mathcal{D}}$ and $\mathbb{P}_{\mathcal{D}'}$ adopt densities $p_{\mathcal{D}}$ and $p_{\mathcal{D}'}$, respectively. Prove that the TV distance is a Wasserstein distance in the sense that

$$\frac{1}{2}\mathrm{TV}(\mathcal{D}, \mathcal{D}') = \min_{\pi \in \Pi(p_{\mathcal{D}}, p_{\mathcal{D}'})} \mathbb{E}_{\pi} d(x, y) \ ,$$

where $d(x, y) = \mathbf{1}_{x \neq y} = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y \end{cases}$.

Problem 6. (30pt)

1. Describe what is the "reparameterization trick" in a variational autoencoder (VAE) and why is it necessary.

2. Let $\boldsymbol{x}^{(i)}$ and $\hat{\boldsymbol{x}}^{(i)}$ denote the input and the decoded reconstruction (i.e. the output) of a VAE. From our discussion in class, the reconstruction error $\left\|\boldsymbol{x}^{(i)} - \hat{\boldsymbol{x}}^{(i)}\right\|_2^2$ corresponds to $\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(i)})}\left[\log p_\theta(\boldsymbol{x}^{(i)}|\boldsymbol{z})\right]$. If we change the reconstruction error to $\left\|\boldsymbol{x}^{(i)} - \hat{\boldsymbol{x}}^{(i)}\right\|_2$, but keep the same variational loss (the KL loss), can we still explain the total loss of the VAE as variational Bayes? If yes, what change do we need to make?

3. Consider a linear autoencoder that takes inputs from $\mathbb{R}^D$. Suppose it has an encoder $\mathbf{P} \in \mathbb{R}^{d\times D}$ and a decoder $\mathbf{Q} \in \mathbb{R}^{D\times d}$, and is trained by minimizing $\sum_i \left\|\boldsymbol{x}^{(i)} - \mathbf{PQ}\boldsymbol{x}^{(i)}\right\|_2$. Suppose the data matrix is full-rank (i.e. the data span the ambient space). Prove: the optimal matrices $\mathbf{P}^*$ and $\mathbf{Q}^*$ satisfy $\mathbf{P}^*\mathbf{Q}^* = \mathbf{A}$ where $\mathbf{A} = \mathbf{A}^\top = \mathbf{A}^2$ (i.e. $\mathbf{A}$ is an orthogonal projection).

–END OF EXAM–